MIL-P-16406A (OS)
9 July 1974
SUPERSEDING
MIL-P-16406 (BuOrd)
5 July 1951

MILITARY SPECIFICATION

PAINT, BLACK, HIGH-GLOSS, QUICK DRYING

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of air-drying phenolic resin coating for use on electrical equipment. This specification meets Air Pollution Regulations (Rule 66).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-S-698	Steel, Sheet and Strip, Low-Carbon
TT-B-846	n-Butyl Alcohol
TT-P-143	Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing and Marking of
TT-T-548	Toluene

FSC 8010

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Military

MIL-L-2104	Lubricating Oil, Internal Combustion Engine, Tactical Service

STANDARDS

Federal

FED-STD-141	Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling and Testing
FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM D115	Methods of Testing Varnishes Used for Electrical Insulation

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 **Color.** The color of the paint shall be Black No. 17038 and shall match the standard color chip of FED-STD-595 when tested as in 4.4.3.

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3.2 **Composition.**

3.2.1 **Vehicle.** The vehicle shall be a phenolic varnish with 12.5 gal. tung oil length. The vehicle shall give a negative test for rosin and rosin derivatives.

3.2.2 **Volatile content.** The volatile content shall conform to the following requirements by volume when tested as in 4.4.4.

a. Solvents having an olefinic or cyclo-olefinic type of unsaturation: 5 percent maximum

b. A combination of aromatic compounds with eight or more carbon atoms to the molecule except ethyl benzene: 8 percent maximum

c. A combination of ethyl benzene, ketones having branched hydrocarbon structures, or toluene: 20 percent maximum

d. Total of a + b + c: 20 percent maximum.

3.3 **Quantitative requirements.** The paint shall conform to the quantitative requirements specified in table I.

Table I

QUANTITATIVE REQUIREMENTS

Characteristics	Minimum	Maximum
Total solids, % by weight of paint	48	–
Specular gloss, 60°	87	–
Drying time:		
Dust free, minutes	–	20
Dry hard, hours	–	1 3/4
Rosin and rosin derivatives	Negative	

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3.4 Quality requirements.

3.4.1 **Condition in container.** When tested as in 4.4.5 the paint shall show little or no settling in the freshly opened package and shall mix readily to a smooth homogeneous state. It shall show no evidence of livering or curdling and shall be free from skins and lumps.

3.4.2 **Brushing properties.** When tested as in 4.4.6 the paint shall brush satisfactorily in all respects and shall dry to a smooth, uniform film free from seeds, runs, sags, or streaks.

3.4.3 **Spraying properties.** When tested as in 4.4.7 the paint shall spray satisfactorily in all respects and shall show no running, sagging, or streaking. The dried film shall show no dusting, mottling, seeding, hazing or other film defects.

3.4.4 **Flexibility.** When tested as in 4.4.9 the paint shall not crack, flake or show loss of adhesion at the bend.

3.4.5 **Immersion resistance.** When tested as in 4.4.10 the paint shall show no wrinkling, blistering, softening, loss of adhesion, or other film defects.

3.4.6 **Carbon tetrachloride resistance.** When tested as in 4.4.11 the paint shall show no wrinkling, blistering, softening, loss of adhesion, or other film defects.

3.4.7 **Dielectric strength.** When tested as in 4.4.12 the dielectric strength of the paint shall not be less than 500 volts/mil dry.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 **Sampling, inspection and testing.** Unless otherwise specified, sampling, inspection, and testing shall be in accordance with method 1031 of FED-STD-141.

4.3 **Classification of tests.** Testing under this specification shall be for the purpose of acceptance of individual lots.

4.4 **Test methods.**

4.4.1 **Test conditions.** The routine and referee testing conditions shall be in accordance with section 7, FED-STD-141 except as otherwise specified herein.

4.4.2 The following tests shall be conducted in accordance with applicable methods of FED-STD-141 or as required in this specification. The right is reserved to make any additional tests deemed necessary to determine that the paint meets the requirements of this specification.

Table II

INDEX

Tests	Test Method		
	Applicable method in FED-STD-141	Paragraph of this specification giving further references	Paragraph of this specification giving requirements
Total solids	4041	–	Table I
Specular gloss, 60°	6101	–	Table I
Drying time	4061	–	Table I
Rosin and rosin derivatives	5031	–	Table I
Color	4250	4.4.3	3.1
Volatile content	7355, 7356	4.4.4	3.2.2
Condition in container	3011	4.4.5	3.4.1
Brushing properties	4321	4.4.6	3.4.2
Spraying properties	4331	4.4.7	3.4.3
Panel preparation	–	4.4.8	–
Flexibility	6221	4.4.9	3.4.4
Immersion resistance	6011	4.4.10	3.4.5
Carbon tetrachloride resistance	–	4.4.11	3.4.6
Dielectric strength	–	4.4.12	3.4.7

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4.4.3 **Color.** In accordance with method 4250 of FED-STD-141 compare the color chip of FED-STD-595 with the pigmented coating prepared in 4.4.8 for compliance with 3.1.

4.4.4 **Volatile content.** Determine solvents as in method 7355 and 7356 of FED-STD-141 for compliance with 3.2.2.

4.4.5 **Condition in container.** Determine package condition in accordance with method 3011 of FED-STD-141 and observe for compliance with 3.4.1.

4.4.6 **Brushing properties.** Apply the packaged paint to a 4- by 12-inch steel panel, using a 1-1/2-inch brush, in accordance with method 4321 of FED-STD-141 for compliance with 3.4.2.

4.4.7 **Spraying properties.** Reduce the paint with thinner conforming to table III in a ratio recommeded by the manufacturer. Spray on a steel panel conforming to QQ-S-698, to a dry film thickness between 0.0009 and 0.0011 inch and observe for spraying properties according to method 4331 of FED-STD-141 for compliance with 3.4.3. For referee test, use automatic application per method 2131 of FED-STD-141.

Table III

THINNER

Ingredient	Percent by weight
VM&P naphtha (8% max. aromatic)	65
n-Butyl alcohol (TT-B-846)	20
Toluene (TT-T-548)	15

4.4.8 **Panel preparation.** Spray the paint as in 4.4.7 on sixteen (16) 3- by 6-inch steel panels solvent cleaned in accordance with method 2011 of FED-STD-141. Air dry for 48 hours and use where applicable in subsequent tests.

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4.4.9 **Flexibility.** Prepare two panels as in 4.4.8 and determine flexibility as in method 6621 of FED-STD-141. Bend over a 3/8-inch mandrel and examine for compliance with 3.4.4.

4.4.10 **Immersion resistance.** Immerse steel panels prepared as in 4.4.8 according to method 6011 of FED-STD-141 in each of the following:

Distilled water	18 hours
Denatured alcohol	30 minutes
Kerosene	30 minutes
Lubricating oil (MIL-L-2104)	30 minutes
6 N hydrochloric acid	30 minutes

For distilled water and 6N hydrochloric acid coat all exposed, uncoated metal surfaces with wax or other suitable coating. Check for compliance with 3.4.5.

4.4.11 **Carbon tetrachloride resistance.** Prepare two panels as in 4.4.8 and rub vigorously for one minute with a rag soaked in carbon tetrachloride. Examine for compliance with 3.4.6.

4.4.12 **Dielectric strength.** Prepare two panels as in 4.4.8 and determine dielectric strength according to ASTM Standard D115 for compliance with 3.4.7.

5. PREPARATION FOR DELIVERY

5.1 The paint shall be furnished in pressurized dispensers, 1-quart or 1-gallon multiple friction top containers, in 5-gallon lug cover steel pails or in 55-gallon steel drums as specified (see 6.2). The paint shall be packaged level A or C; packed level A or C as specified (see 6.2) in accordance with TT-P-143.

5.2 **Marking.** In addition to any special marking interior packages and exterior containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 The paint covered by this specification is intended for use primarily as an insulating protective coating on electrical equipment. It is also useful as a oil resistant protective coating and sealant.

6.2 **Ordering data.** Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Size of container.
c. Level of packaging and level of packing.

6.3 The paint should be purchased by volume, the unit being one U.S. liquid gallon of 231 cubic inches at 68°F (20° C).

6.4 Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:

Navy-OS

Preparing activity:

Navy-OS
(Project No. 8010-N115)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-P-16406A (OS), PAINT, BLACK, HIGH-GLOSS, QUICK DRYING

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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